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Supplementary Material: Challenges of Gate-Based Quantum Optimization for a Discrete Process-Design Problem

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ABSTRACT

This supplement records provenance, validation, and artifact anchors for the main manuscript on gate-based quantum optimization of a drug-substance manufacturing (DS-MFG) discrete process-design problem. It is intended to be compiled and submitted separately from the main conference paper if appendices are not counted as article body material in the submission workflow.

Keywords: supplementary material, quantum optimization, process design, reproducibility

REPOSITORY AND PACKAGING

The main manuscript keeps only the narrative-critical values. This supplement keeps the audit data close to the manuscript source while allowing the main article to remain within the conference page limit. The repository URL is https://github.com/SECQUOIA/pse-quantum-discrete-optimization. A final archival DOI has not yet been assigned. It will be added before camera-ready submission if one is minted; the review artifact is identified by the repository snapshot that contains this supplement and by the checksums below.

Table 1: Current reproducibility pins validated for the DS-MFG case study.

Layer	Current pin or check	Repository anchor
Julia optimization wrapper	`QiskitOpt.jl` v0.7.0; manifest git tree `a601dccdd43fd0a8b18e63f3b771273817af49eb`.	`case-studies/ds-mfg-qubo-qiskitopt/Project.toml`; `case-studies/ds-mfg-qubo-qiskitopt/Manifest.toml`.
Julia QUBO stack	`QUBODrivers.jl` v0.6.1, `QUBOTools.jl` v0.13.0, `QUBO.jl` v0.6.0.	`case-studies/ds-mfg-qubo-qiskitopt/Project.toml`; `case-studies/ds-mfg-qubo-qiskitopt/Manifest.toml`.
Python bridge	Qiskit 2.3.1, Qiskit Aer 0.17.2, Qiskit Optimization 0.7.0, Qiskit IBM Runtime 0.46.1, NumPy 2.2.6, SciPy 1.15.3.	`case-studies/ds-mfg-qubo-qiskitopt/CondaPkg.toml`.
Fresh-clone validation	`julia --project=. scripts/smoke_test.jl` imports packages, checks required artifacts, and validates cached summary values.	`case-studies/ds-mfg-qubo-qiskitopt/scripts/smoke_test.jl`.

Table 2: Reproducibility levels used for the DS-MFG manuscript package.

Level	Meaning	Current support
Score-from-cache reproducible	Retained samples, summaries, and repair tables can be inspected and rescored without rerunning the stochastic campaigns.	Supported by cached distributions, hit-rate reports, checksums, and the smoke test. This is the main reproducibility level for the manuscript claims.
Rerunnable with retained scripts	The scripts and pinned environment can rerun selected QAOA/VQE, classical, and hardware-handoff workflows, subject to stochastic variation and credential availability.	Partially supported. QAOA fixed-angle parameters and sampled outputs are retained; selected VQE metadata reruns now retain optimized ansatz vectors for seed 74018 and for the final-budget selected seeds 74018, 74007, and 74001.
Full campaign replay	Every stochastic campaign can be reproduced byte-for-byte from retained optimizer states, simulator logs, hardware calibration, and submitted-circuit metadata.	Not claimed. Some optimizer, MPS truncation, and submitted-job hardware metadata were not retained.

Table 3: Checksums for retained archives and primary cached summaries.

Artifact	SHA-256 checksum
`case-studies/ds-mfg-qubo-qiskitopt/Fw_DSmfgcasequboinformation.zip`	`aa20d793fc45c25c2bd7cc0a231877f71b225bf80daee11531cef6eb6cac5af2`
`ds_mfg_ibm_qaoa_pilot_fez_4096x3.zip`	`a3f5441d81a107ff99e7df3a1ab4b0e1ef4df6de813ed3416d4997d1d6262b83`
`ds_mfg_direct_full_qubo_hardware_pilot.zip`	`25a39b061f588523eeb609c397abc3b2afd8dd4f15c4f9d6d8037772532c6515`
`case-studies/ds-mfg-qubo-qiskitopt/ds_mfg_qaoa_juliqaoa_transfer_highread/qaoa_juliqaoa_transfer_summary.csv`	`10bb3bc9b746fb12a87eadc1fd2b819c1a3cacbc84025c9bd54797c1088c34b7`
`case-studies/ds-mfg-qubo-qiskitopt/ds_mfg_vqe_reduced_flow_objective_final/vqe_reduced_top50_sampling_summary.csv`	`aac0299d49314b8d0e4d54b088f50cc728f1d212540613348ab62d209a132ee1`
`case-studies/ds-mfg-qubo-qiskitopt/ds_mfg_gurobi_provenance/ip_exact_enumeration_summary.json`	`1298dcb07aa82c0adc05ead3fb1b3f4b86ae9ef38e9a276b98f97ec5a7606e5e`
`case-studies/ds-mfg-qubo-qiskitopt/ds_mfg_gurobi_provenance/gurobi_local_pool_rerun_summary.json`	`13f7683c41cbf14b4225326e41778c6f42ae3979e4d1303af65c7132fe35a6e7`
`case-studies/ds-mfg-qubo-qiskitopt/ds_mfg_vqe_reduced_flow_objective_selected_metadata/vqe_reduced_top50_sampling_summary.csv`	`df82de49b921e3b70d5d24f0896e77b6da64477c3a13cffc0a0761d6e9ebd2db`
`case-studies/ds-mfg-qubo-qiskitopt/ds_mfg_hit_rate_reports/time_to_solution_report.csv`	`c27e70684f3b753efc4eb8cf2b75e79afe4b15402bcd6b3ca7d7a598310f2efd`

VALIDATION FACTS

Table 4: Numerical validation facts used to support the manuscript claims.

Check	Retained value	Anchor
Exact repaired optimum	$F(y^{\star}) = 11.7095$ at flow `1001110100111100011`.	`ds_mfg_reduced_flow_objective/reduced_flow_summary.csv`; `ds_mfg_reduced_flow_objective/reduced_exact_top_flows.csv`.
Gurobi pool bridge	All 36 stored Gurobi-pool flows match the repaired QUBO score within 3.1×10^{-13}.	`ds_mfg_reduced_flow_objective/reduced_exact_top_flows.csv`; `ds_mfg_gurobi_provenance/gurobi_local_pool_rerun_solutions.csv`.
Retained Gurobi metadata	Model `capex`: 19 binary variables, 20 constraints, 58 nonzeros; `PoolSearchMode=2`, `PoolSolutions=36`, solve time 0.073 s, 36 solutions found. Solver version, status code, MIPGap, PoolGap, and pool completeness/diversity intent were not retained in the original export. A local Gurobi 13.0.2 rerun with `PoolSearchMode=2` and `PoolSolutions=100` returned `OPTIMAL`, `MIPGap=0.0`, 36 solutions, and exact retained-pool agreement.	`Fw_DSmfgcasequboinformation.zip/gurobi_pool_solutions_n36_20260126_105706.json`.
Quadratic surrogate weakness	Global fit $R^2 \approx 0.999$, but the surrogate optimum repairs to 14.5505. The true repaired optimum is surrogate rank 33; the surrogate optimum is exact repaired rank 6.	`ds_mfg_reduced_flow_objective/reduced_flow_summary.csv`; Figure 1.
Low-energy surrogate audit	Exact enumeration finds 36 feasible flows (the complete pool). The 50 lowest-cost repaired flows include 14 infeasible flows — the penalty weights are insufficient to push all infeasible flows above all feasible ones. In the lowest 1% repaired-flow set, exact-rank versus surrogate-rank Spearman correlation is 0.901. Exact enumeration wall time was not retained.	`ds_mfg_reduced_flow_objective/reduced_exact_top_flows.csv`; `MANUSCRIPT_FINDINGS.tex`.
Primary QAOA simulator check	JuliQAOA statevector probability on the reference optimum was 2.63×10^{-3} for the energy-targeted $p = 5$ angles. The cached Aer MPS sampled rate was 2.53×10^{-3}, inside its Wilson 95% interval $[2.35, 2.73] \times 10^{-3}$. MPS truncation logs were not retained.	`ds_mfg_qaoa_juliqaoa_angle_search/juliqaoa_angle_summary.csv`; `ds_mfg_qaoa_juliqaoa_transfer_highread/qaoa_juliqaoa_transfer_summary.csv`.

Table 5: Implications of missing provenance values for the manuscript claims.

Missing value	Affects main claim?	Affects full rerun?	Current mitigation
Historical Gurobi final status and MIPGap	No for the current claim after rerun; the original export still lacks these fields, but the model/pool behavior has been rerun locally.	Partly; exact historical solver state is unavailable, but current solver provenance is retained.	The same flow is certified by exact enumeration over all 2^{19} flows and by a local Gurobi 13.0.2 rerun with `OPTIMAL` status, `MIPGap=0.0`, 36 solutions, and exact retained-pool agreement.
MPS truncation logs	No for the central QAOA sampling claim.	Yes, for simulator-forensics replay.	The energy-targeted QAOA row is cross-checked against independent JuliQAOA statevector probabilities, which agree with the Aer MPS sampled rate within the Wilson interval.
VQE optimized parameter vectors	No; VQE is secondary and reported as a retained sampled distribution.	Partly; selected fixed-circuit replay is now supported, but full historical optimizer replay is not claimed.	Metadata reruns retain optimized `EfficientSU2` vectors for seed 74018 and for seeds 74018, 74007, and 74001 at the final sampling budget. The current metadata rerun produced 4 reference hits total and is provenance, not a replacement for the historical 28-hit final follow-up.
Submitted hardware layout, depth, and two-qubit count	No, because the hardware row is a submission-and-scoring handoff only.	Yes, for hardware-performance diagnosis.	The manifest retains backend, job IDs, shots, seeds, logical circuit metadata, and scored counts. No noise-mechanism claim is made.
Hardware calibration or mitigation metadata	No under the current claim.	Yes, for calibrated hardware analysis.	The manuscript states that no readout or error mitigation is claimed and that the retained metadata cannot diagnose a specific noise source.
Exact enumeration wall time	No.	Partly, for cost accounting.	The enumeration is documented as exact over all 2^{19} flows; issue 24 tracks recording wall time if the enumeration script is rerun.

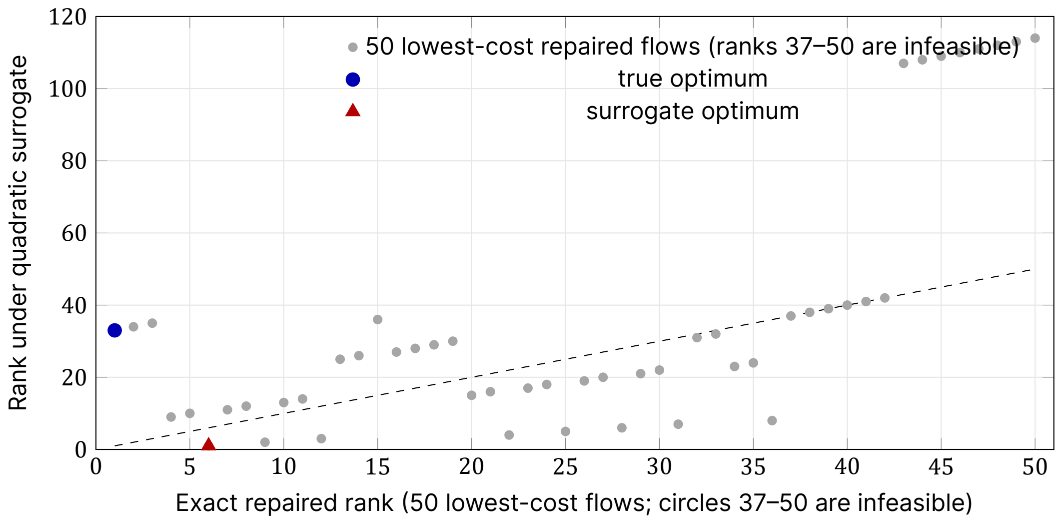

Figure 1. Surrogate rank audit for the 50 lowest-cost repaired flows. Exact enumeration confirms there are exactly 36 feasible flows (ranks 1–36); the remaining 14 shown here are infeasible flows whose penalty values are insufficient to place them above all feasible ones. The surrogate has high global R^2 but imperfect low-energy ordering: the exact optimum is only rank 33 under the surrogate, while the surrogate minimum repairs to the sixth-best exact flow. Quantum-sampling outcomes are therefore scored using the exact repaired objective F, not the surrogate $\widehat{F}$.

VARIATIONAL AND HARDWARE PROVENANCE

Table 6: Variational-parameter provenance for the central QAOA and VQE rows.

Workflow	Circuit / ansatz	Objective and optimizer	Parameter source	Validation metric
Energy-targeted QAOA	$p = 5$ QAOA on the 19-flow surrogate.	Minimize $\mathbb{E}_{p_\theta}[\hat{F}(y)]$ with JuliQAOA `find_angles_bh`; seed 91001; 5 basin-hopping iterations.	`ds_mfg_qaoa_juliqaoa_angle_search/juliqaoa_angle_summary.csv`; angles transferred beta-then-gamma to `QiskitOpt.QAOA`.	664 reference hits / 262144 reads; 39661 top-50 reads.
Top-10 QAOA diagnostic	$p = 5$ QAOA on the 19-flow surrogate.	Maximize probability on the enumerated exact top-10 set; oracle-aided because the repaired ranking is known.	Top-10 angle vector retained in `ds_mfg_ibm_qaoa_pilot_fez_4096x3.zip/job_manifest.json`; local 1007-hit diagnostic CSV was not retained.	1007 reference hits / 262144 reads in the retained manuscript summary; hardware handoff used this circuit.
Reduced-surrogate VQE	`EfficientSU2`; 152 initial parameters on the 19-flow surrogate.	COBYLA through `QiskitOpt.VQE`; 128 optimizer reads; 25 iterations; seeds 74001, 74007, and 74018 in the final follow-up.	Historical sampled distributions retained; current metadata reruns persist optimized ansatz vectors in `ds_mfg_vqe_reduced_flow_objective_seed74018_metadata/` and `ds_mfg_vqe_reduced_flow_objective_selected_metadata/`.	Historical final follow-up: seed 74018 had 20 reference hits / 524288 reads; three seeds had 28 reference hits / 1572864 reads. Current metadata rerun: 4 reference hits / 1572864 reads.

Table 7: Hardware pilot provenance retained in the submitted-job archive.

Item	Retained metadata	Limitation
Backend and jobs	`ibm_fez`; backend queried 2026-06-17 20:19 UTC; 156 qubits; 9 completed jobs; 4096 shots per job; repeats 1–3; transpile seeds 92001–92003; job IDs retained.	Queue time, calibration snapshot, readout mitigation, and error mitigation metadata were not retained.
Logical circuit	19 qubits and 19 classical bits; pre-transpile depth 86; operation counts: 19 `h`, 95 `rx`, 95 `rz`, 255 `rzz`, and 19 measurements; scoring reverses Qiskit count keys to $x_1, \ldots, x_{19}$.	Submitted-job transpiled depth, two-qubit-gate count, physical layout, and measurement map were not retained.
Hardware outcome	36864 total reads; 6 top-50 reads; 1 top-10 read; 4 stored Gurobi-pool reads; 0 reference-optimum reads; best repaired flow rank 2 with objective 11.8105. With zero reference hits, the two-sided 95% Wilson upper bound on the reference-optimum rate is 1.04×10^{-4}.	The pilot validates submission and scoring only; it cannot diagnose a specific noise mechanism.